EXPERIMENT 5: WIRESHARK PACKET CAPTURE

AIM

To study Wireshark as a packet capture and network analysis tool and understand the encapsulation of information at different layers of a protocol stack.

INTRODUCTION

Wireshark is a network protocol analyser that captures packets in real time and displays their contents in a human-readable format. It helps examine network traffic, inspect protocol fields, and troubleshoot communication problems. A packet sniffer is passive: it observes packets rather than generating network traffic itself.

PROCEDURE

1. Open Wireshark and select the active network interface, such as Ethernet or Wi-Fi.
2. Start capturing packets and generate network traffic by opening a website or using the ping command.
3. Observe captured packets in the Packet List pane and select a packet to inspect its protocol fields.
4. Expand protocol layers in Packet Details and examine the raw hexadecimal data in Packet Bytes.
5. Stop the capture and apply a display filter when required.

OBSERVATIONS

• Wireshark captures packets sent and received through the selected network interface.
• Protocol fields can be expanded and inspected at different layers.
• Display filters isolate traffic such as DNS, TCP, UDP, or ICMP.
• Colour coding helps distinguish traffic types and identify packets that may require attention.

ENCAPSULATION

Protocol layer	Information observed
Application	Application data, such as a DNS query or HTTP request
Transport	TCP or UDP header, including source and destination ports
Network	IP header containing source and destination IP addresses
Data Link	Ethernet header containing source and destination MAC addresses

STUDENT OBSERVATION

1. What is promiscuous mode?
Promiscuous mode allows a network interface to capture frames it can observe, including frames not addressed to its own MAC address, depending on the network and adapter.

2. Do ARP packets have a transport-layer header? Explain.
No. ARP is carried directly in an Ethernet frame and does not use TCP or UDP.

3. Which transport-layer protocol is used by DNS?
DNS commonly uses UDP port 53; TCP port 53 is also used in cases such as zone transfers and some large responses.

4. What is the port number used by HTTP?
HTTP uses TCP port 80 by default.

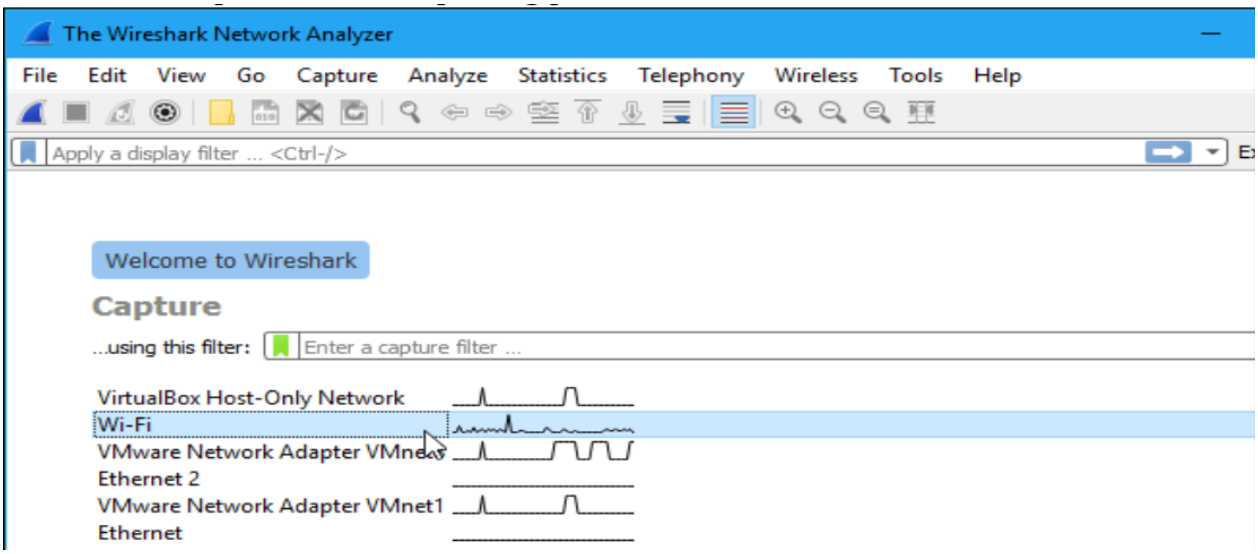
5. What is a broadcast IP address?
It is an address used to send a packet to all hosts on a particular IP network; for example, 192.168.1.255 is the directed broadcast address for 192.168.1.0/24.

RESULT

Wireshark was studied as a packet capture tool. Packets were examined through the Packet List, Packet Details, and Packet Bytes panes, demonstrating how protocol information is organised across network layers.

WIRESHARK SCREENSHOTS

1. Selecting a network interface for packet capture



2. Captured packets and protocol details

